

MARSHALL AAF, KS, USA

WMO: 724550

Lat: 39.050N

Lon: 96.767W

Elev: 325

StdP: 97.49

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-13.0	-21.0	0.6	-13.6	-18.4	0.8	-11.5	11.1	5.6	10.0	3.9	2.7	350	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	37.8	24.0	35.7	24.1	34.1	23.8	26.1	33.0	25.4	32.4	24.8	31.7	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.1	29.2	23.6	19.1	28.6	22.9	18.3	28.1	82.8	32.9	79.8	32.4	76.9	31.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.4	-20.1	40.0	2.8	1.5	-22.1	41.1	-23.7	42.0	-25.3	42.8	-27.3	44.0		
(5)			-20.4	27.4	2.6	0.9	-22.3	28.1	-23.9	28.6	-25.3	29.2	-27.3	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-1.0	1.0	8.1	13.0	18.6	24.8	26.9	25.9	21.4	14.0	7.3	0.4
	DBStd	11.11	6.02	6.76	6.42	5.31	4.70	3.56	3.44	3.51	4.74	5.28	5.45	5.87
	HDD10.0	1182	343	261	112	29	2	0	0	0	0	21	113	301
	HDD18.3	2703	601	485	322	173	57	1	0	1	22	153	333	557
	CDD10.0	2435	1	9	53	119	269	444	523	494	342	146	31	3
	CDD18.3	914	0	0	4	13	66	196	264	236	115	19	1	0
	CDH23.3	10373	0	4	62	170	676	2202	3171	2665	1196	215	11	0
	CDH26.7	4884	0	0	10	42	226	1021	1675	1347	506	57	1	0
Wind	WSAvg	3.6	3.3	3.7	4.2	4.5	3.8	3.9	3.3	3.2	3.5	3.5	3.7	3.3
Precipitation	PrecAvg	824	17	27	55	76	117	127	119	99	72	57	35	24
	PrecMax	1272	39	71	116	213	339	234	355	164	127	139	95	77
	PrecMin	590	1	0	0	22	29	23	40	14	8	5	0	1
	PrecStd	153	10	19	30	41	82	50	79	40	34	34	26	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	23.4	27.9	30.5	33.6	37.8	40.5	39.6	36.4	31.3	24.9	18.6
		MCWB	10.9	11.9	15.8	18.5	21.2	23.4	23.5	24.1	23.3	20.4	15.3	12.9
	2%	DB	14.5	19.1	25.1	27.6	31.2	35.5	38.3	37.2	33.9	28.1	21.8	15.4
		MCWB	8.2	10.8	14.8	17.2	20.6	24.0	23.9	24.3	23.3	18.3	14.3	9.5
	5%	DB	11.3	15.7	22.2	25.1	29.2	33.9	36.1	35.2	32.0	25.6	19.0	12.6
		MCWB	6.2	10.0	13.2	16.2	20.0	23.8	24.2	24.0	22.8	17.4	12.6	7.8
	10%	DB	8.5	12.5	19.4	22.7	27.4	32.2	34.1	33.2	29.8	23.4	16.4	9.8
		MCWB	4.6	7.4	12.2	15.0	19.4	23.4	24.1	23.6	21.8	16.4	11.0	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	14.9	18.7	20.2	23.5	26.4	27.0	27.3	25.6	22.4	17.3	14.8
		MCDB	14.6	18.2	25.0	26.1	29.3	33.7	33.7	34.8	32.1	27.6	22.7	17.1
	2%	WB	8.6	12.7	16.7	18.9	22.3	25.5	26.1	26.0	24.6	20.9	15.3	11.0
		MCDB	13.3	17.0	21.5	25.0	28.2	32.2	32.2	32.7	31.3	25.1	18.9	13.1
	5%	WB	6.6	10.1	14.8	17.7	21.4	24.7	25.4	25.3	23.7	19.3	13.6	8.1
		MCDB	10.8	16.7	20.8	23.3	27.1	31.5	32.6	32.2	30.1	23.6	18.3	11.9
	10%	WB	4.8	6.9	12.7	16.1	20.5	23.9	24.8	24.6	22.7	17.4	11.5	6.1
		MCDB	8.6	11.9	18.2	21.4	25.8	30.6	31.8	31.2	28.6	21.6	16.1	9.9
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.5	13.5	13.0	11.9	11.6	11.9	11.9	12.3	13.5	12.7	11.3
		MCDBR	17.1	18.3	18.3	16.4	14.2	13.2	14.2	14.3	13.4	16.0	16.0	15.7
		MCWBR	11.3	11.4	9.6	8.7	6.1	4.9	4.4	5.0	5.5	8.2	9.2	10.6
	5% WB	MCDBR	15.9	17.5	14.9	14.0	12.1	11.8	12.0	12.1	11.9	13.4	13.5	14.1
		MCWBR	10.9	11.6	8.6	8.5	6.0	5.4	4.9	5.1	5.4	8.1	9.0	10.6
Clear-Sky Solar Irradiance	taub		0.287	0.302	0.333	0.379	0.411	0.422	0.428	0.427	0.383	0.342	0.302	0.282
	taud		2.487	2.453	2.410	2.306	2.287	2.290	2.298	2.298	2.374	2.465	2.509	2.502
	Ebn at Noon		897	927	926	897	871	858	849	841	860	866	870	875
	Edh at Noon		77	92	107	126	132	131	129	126	110	90	76	71
All-Sky Solar Radiation	RadAvg		2.34	3.10	4.07	5.16	5.89	6.68	6.58	5.87	4.85	3.55	2.56	2.00
	RadStd		0.20	0.32	0.30	0.32	0.47	0.39	0.45	0.30	0.31	0.37	0.21	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page